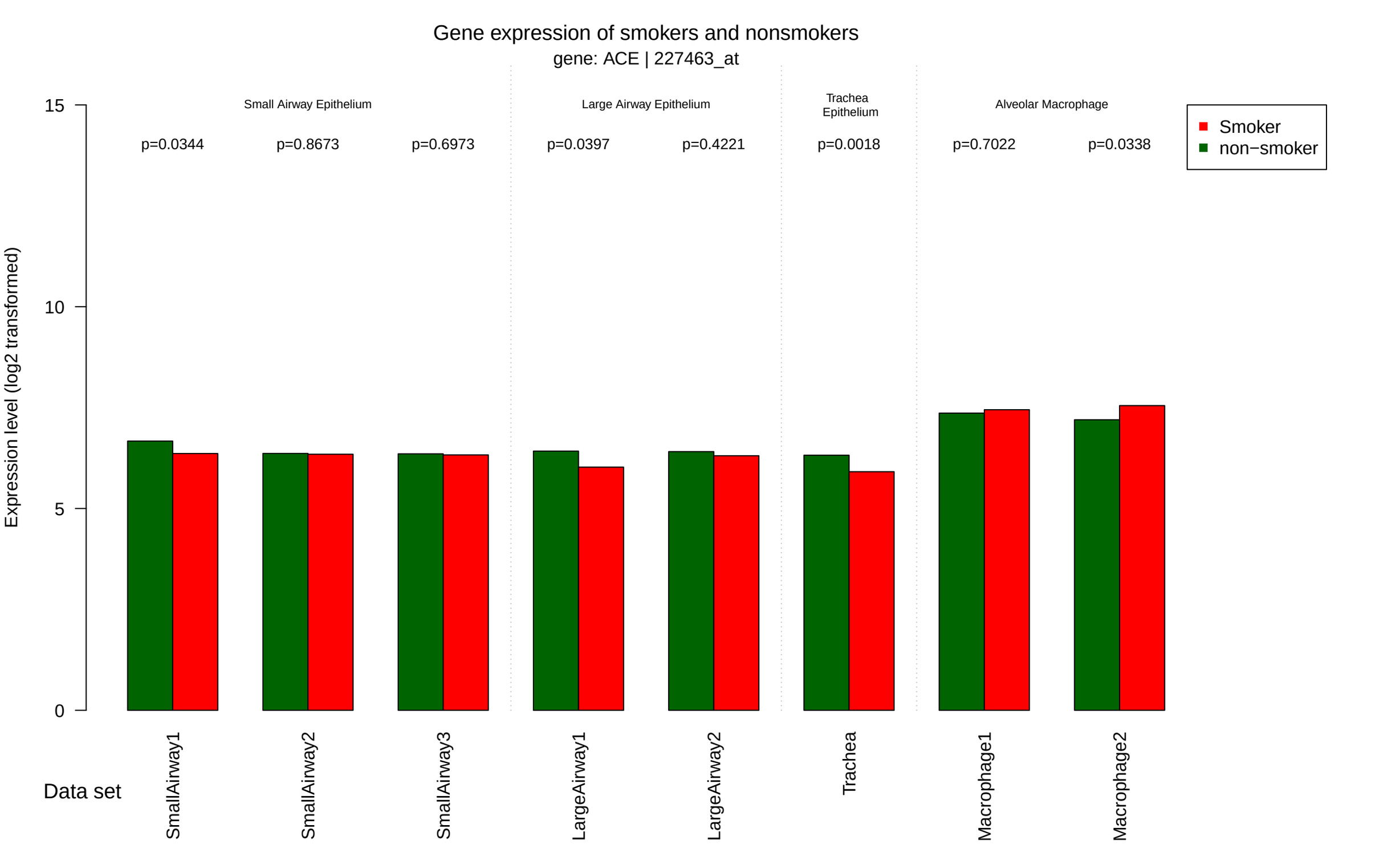
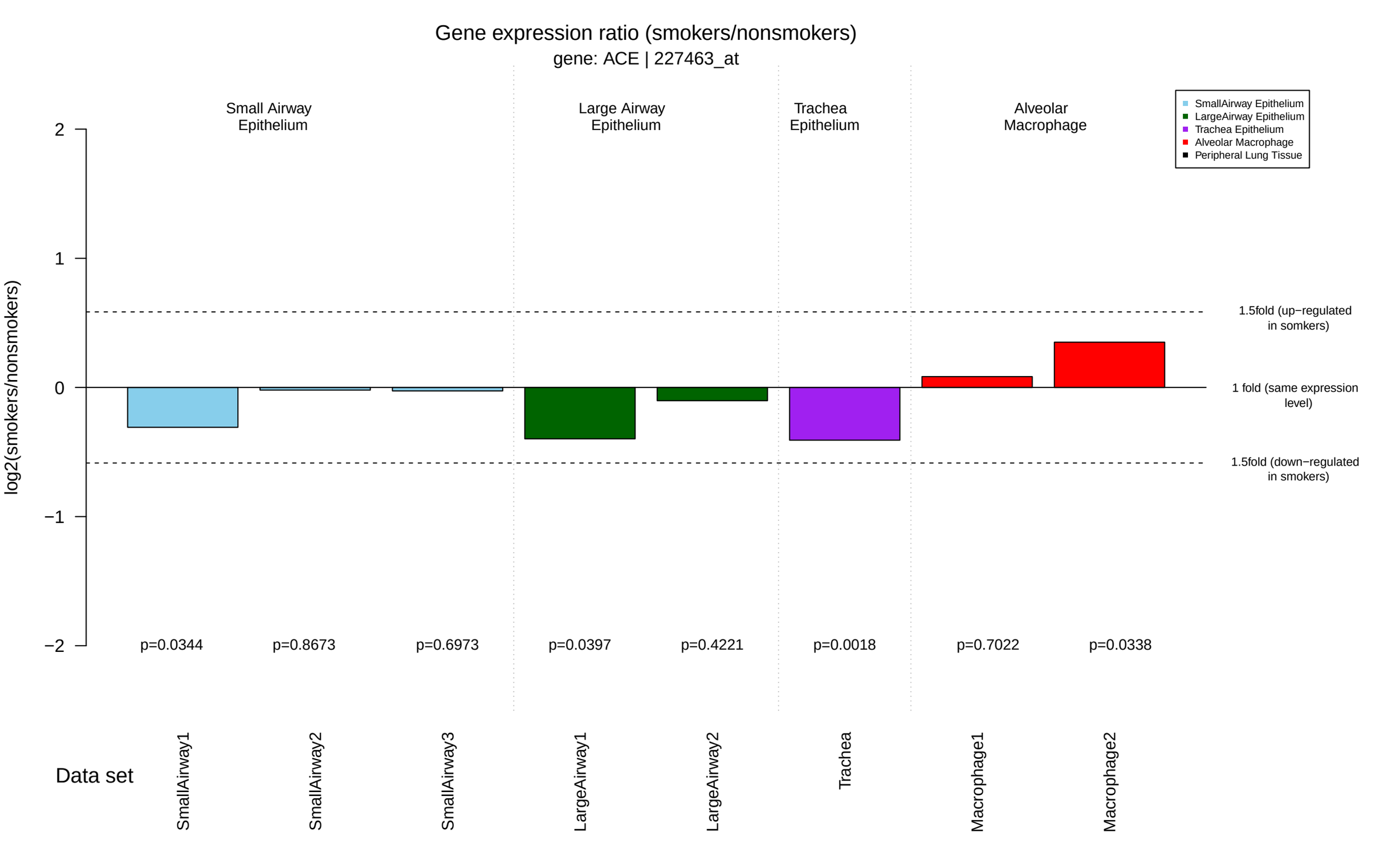
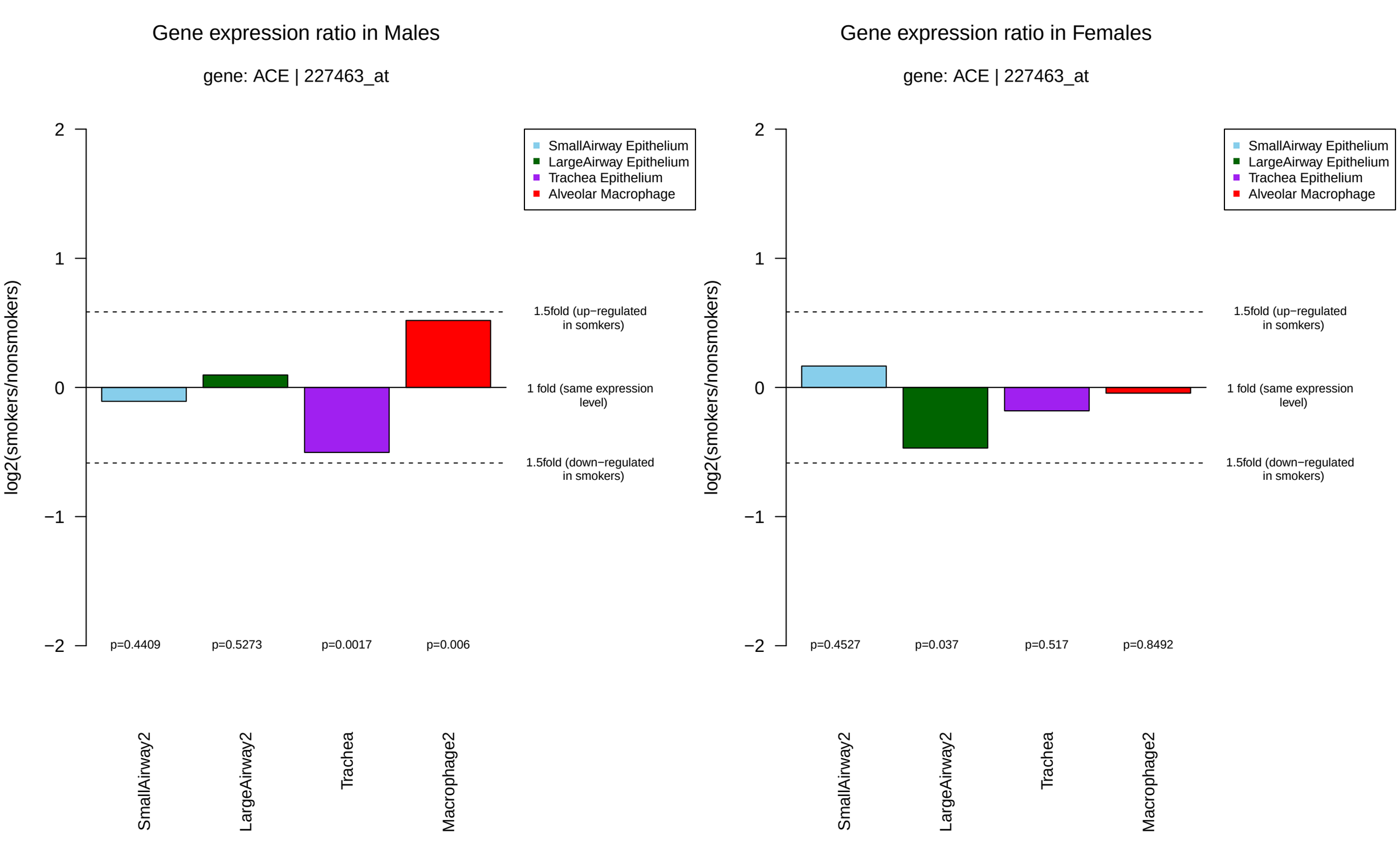
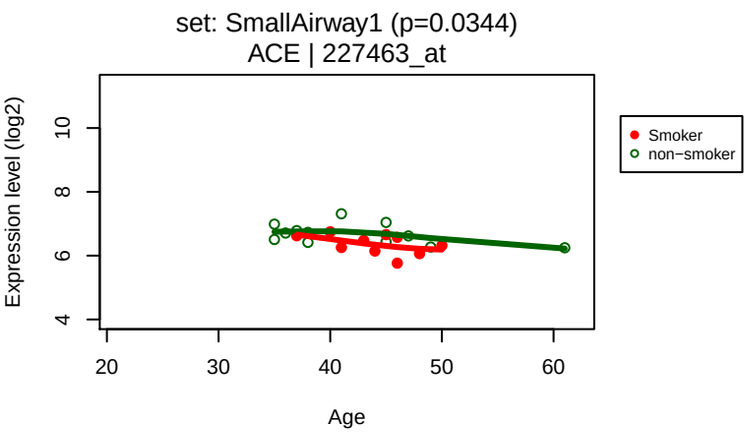
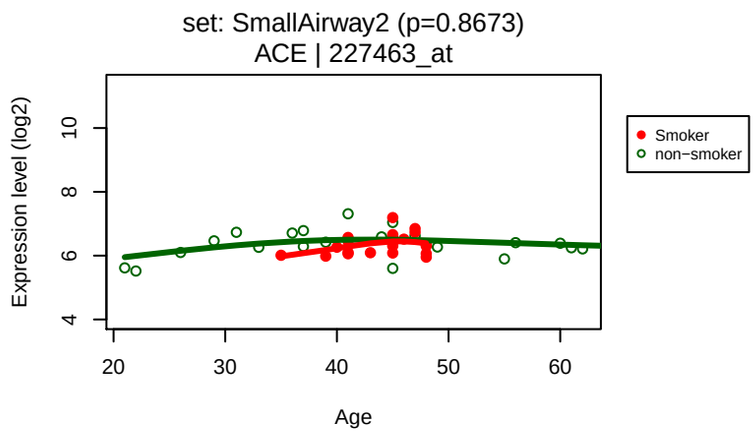
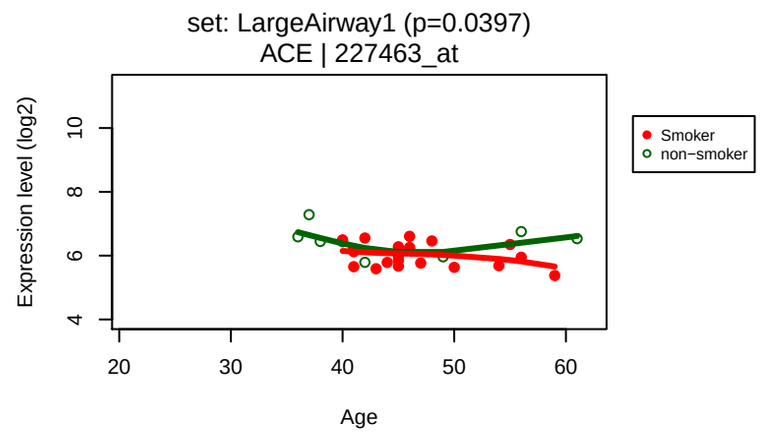

Gene expression of smokers and nonsmokers
gene: ACE | 227463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0344
p=0.8673
p=0.6973
p=0.0397
p=0.4221
p=0.0018
p=0.7022
p=0.0338
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

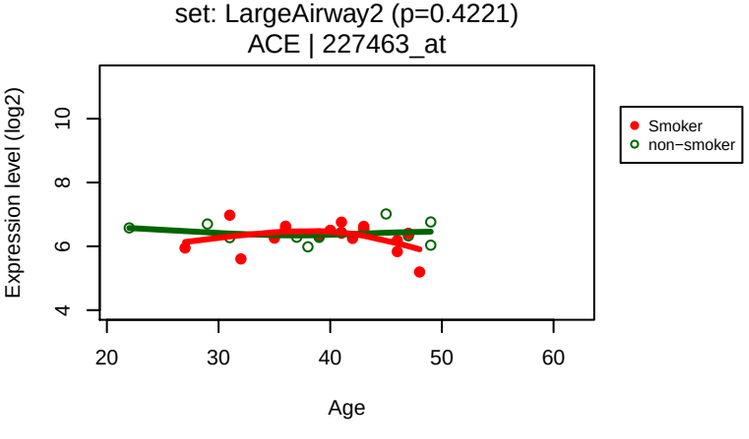

Gene expression ratio (smokers/nonsmokers)
gene: ACE | 227463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0344
p=0.8673
p=0.6973
p=0.0397
p=0.4221
p=0.0018
p=0.7022
p=0.0338
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

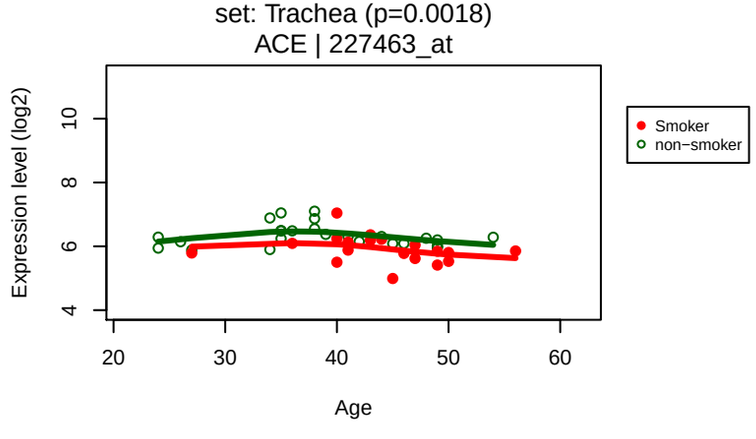

Gene expression ratio in Males
gene: ACE | 227463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4409
p=0.5273
p=0.0017
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACE | 227463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4527
p=0.037
p=0.517
p=0.8492
SmallAirway2
LargeAirway2
Trachea
Macrophage2

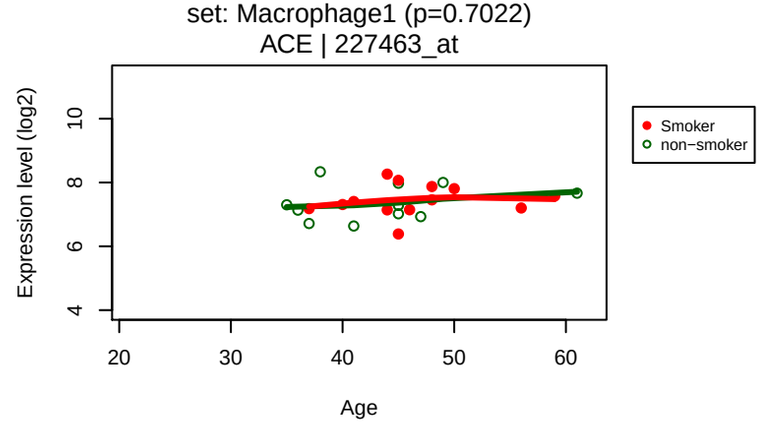

set: SmallAirway1 (p=0.0344)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker

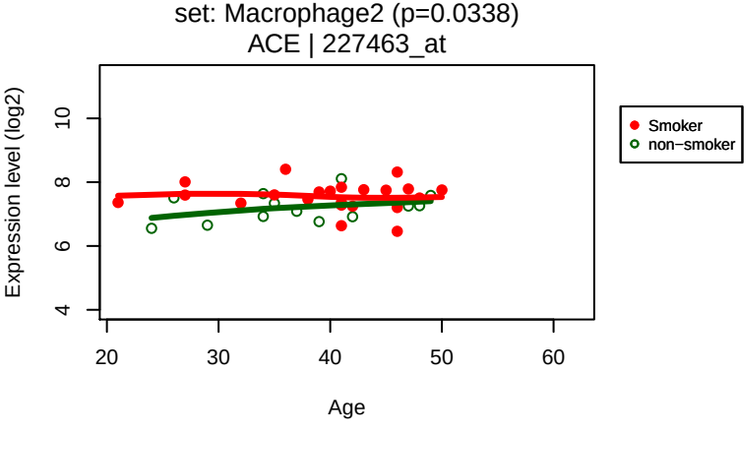

set: SmallAirway2 (p=0.8673)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker

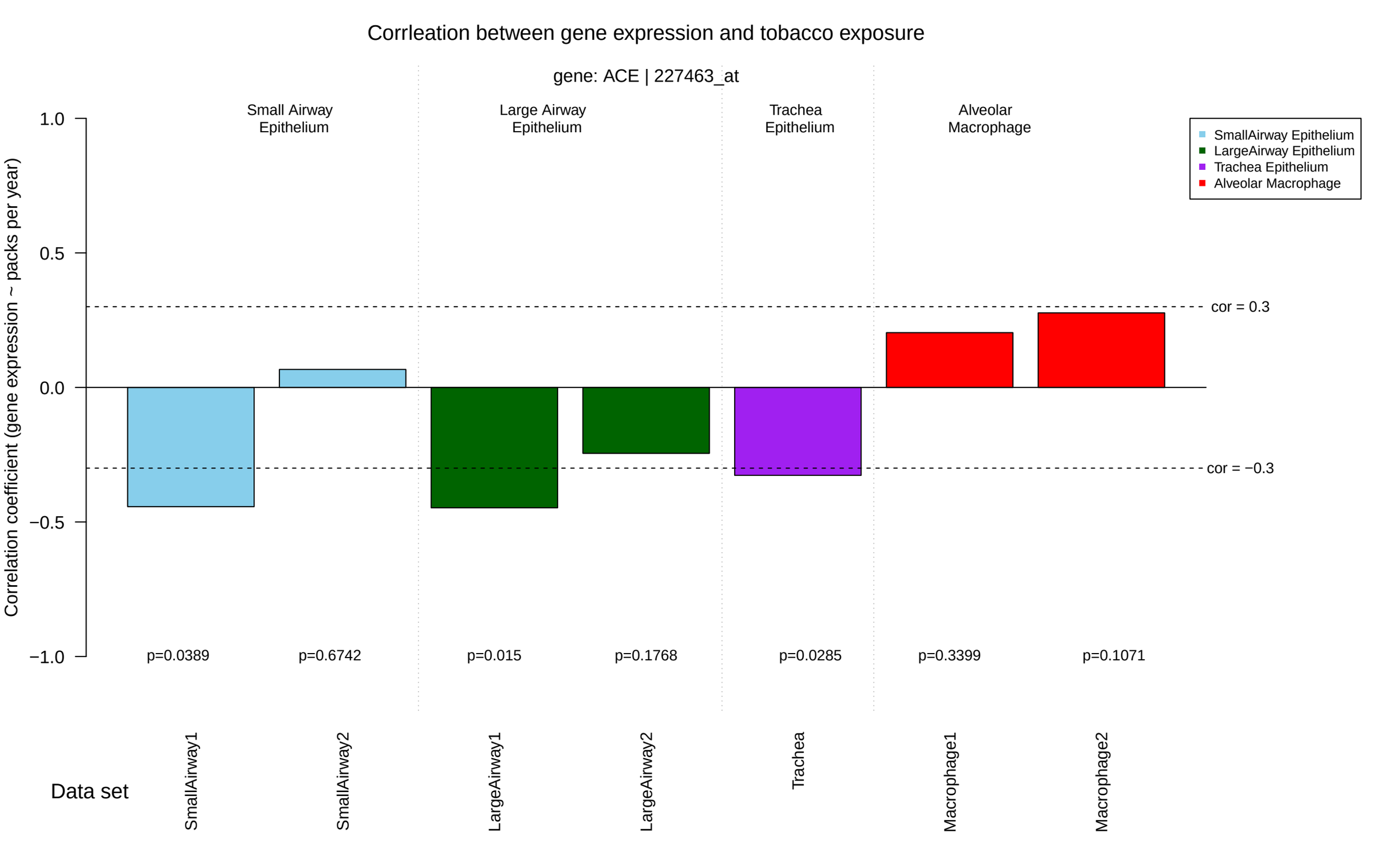

set: LargeAirway1 (p=0.0397)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4221)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7022)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0338)
ACE | 227463_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACE | 227463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0389
p=0.6742
p=0.015
p=0.1768
p=0.0285
p=0.3399
p=0.1071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2